

Checkpoint - Heating Copper Evidence Analysis

Page 1 of 2

Complete every numbered item and submit this PDF.

Student name:

Scenario evidence

Copper begins shiny orange-brown, ends with a black solid coating, and increases from 2.40 g to 2.69 g.

Scraping the coating does not completely restore the original surface.

1. TESTABLE RESEARCH QUESTION

2. VARIABLES

Independent variable

3. TWO SAFETY PRECAUTIONS

4. BEFORE-AND-AFTER EVIDENCE TABLE

Condition	Mass (g)	Color / surface	Other evidence
Before heating			
After heating			

5. CLASSIFICATION DECISION

Does the evidence support a physical change or a chemical change? State your decision.

Heating Copper - CER and Particle Explanation

Page 2 of 2

Use the scenario and your evidence table.

6A. CLAIM

6B. EVIDENCE - AT LEAST THREE SPECIFIC RESULTS

6C. REASONING

7. PARTICLE-LEVEL EXPLANATION

Explain what the black coating and mass increase suggest about copper particles and particles from the air.

FINAL CHECK / CORRECTION RECORD

Confirm all seven items are complete. Record any correction below without erasing the original response.